

KAI FRAZIER

✉ kairfrazier@gmail.com ☎ (425) 970-0000 📍 Seattle, WA 🇺🇸 United States Citizen

WORK EXPERIENCE

Student Front Desk Assistant

07/2026 – Present

UW Department of Electrical & Computer Engineering

- Manage intake, triage, and distribution of laboratory shipments, freight, and technical departmental hardware across engineering facilities.
- Administer enterprise physical security systems, managing keycard access provisioning for laboratory facilities and conference rooms.

Technical & Operational Support Intern

08/2023 – 10/2024

Bureau of Alcohol, Tobacco, Firearms and Explosives (ATF)

- Provisioned and configured mobile communication hardware for agency personnel while maintaining compliance with federal security protocols.
- Handled secure data processing and document chain-of-custody protocols for sensitive digital media and agency records.

EDUCATION

Bachelor of Science in Electrical and Computer Engineering

09/2025 – 03/2028

University of Washington

Seattle, WA

Relevant Coursework: Semiconductor Devices (EE 331), Circuit Theory (EE 233), Digital Circuits & Systems (EE 271), Computer Hardware Skills (EE 201), Signal Processing (EE 241), Advanced Technical Communication (EE 393)

TECHNICAL SKILLS

Hardware & EDA Tools

Ansys HFSS, Ansys Electronics Desktop, SMath, PCB Layout, Schematic Capture, Microstrip Synthesis

Lab Equipment

Digital Storage Oscilloscopes, Vector Network Analyzers (VNA), Digital Multimeters (DMM), Function Generators, DC Power Supplies, Soldering & Rework

Programming & Systems

Python, Fortran, Git/GitHub

ENGINEERING PROJECTS

1090 MHz Parallel-Coupled Microstrip Bandpass Filter

Tools: Ansys HFSS, SMath, Python, Fortran

- Designed and synthesized a 1090 MHz coupled microstrip bandpass filter for an ADS-B aerospace receiver using Chebyshev prototype low-pass parameters and J-inverters.
- Modeled 3D electromagnetic structures and substrate geometries in Ansys Electronics Desktop (HFSS) to validate S-parameters (S11, S21) and passband transmission.
- Developed custom calculation scripts to automate transmission line impedance calculations and physical strip dimensions.

Adjustable AC-to-DC Linear Power Supply

Tools: Hardware Design, Linear Regulators, Lab Prototyping

- Designed and built a transformer-fed, full-wave rectified AC-to-DC power converter converting 120V AC line voltage into a variable, low-noise DC output.
- Integrated smoothing capacitors and adjustable linear regulation stages to minimize output ripple across variable resistive loads.
- Measured voltage regulation, stability, and transient behavior using digital multimeters, DC power supplies, and bench oscilloscopes.

Custom Mixed-Signal PCB Prototype (EE 201)

Tools: EDA CAD, Soldering, Hardware Debugging

- Designed schematic capture and physical trace routing for a mixed-signal printed circuit board adhering to standard board fabrication constraints.
- Populated circuit boards using precision through-hole and surface-mount (SMD) bench soldering techniques.
- Verified signal integrity and board-level functionality using digital storage oscilloscopes, function generators, and DMMs.